

A general forecasting model of patient timelines for clinical alerting and interaction

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Abstract

Most clinical machine-learning models are built for one task at a time, such as mortality or sepsis, and each new alert requires a new model. We describe a single sequence model that forecasts a patient’s whole electronic health record timeline: which events come next, when, and the risk of clinically important events over time. Trained on [TBD] admissions from MIMIC-IV and [TBD] intensive-care stays from eICU-CRD, the model produces calibrated time-to-event risk for vasopressor initiation, acute kidney injury and death from a single pass over the record, matching or exceeding task-specific gradient-boosted models on [TBD] of [TBD] event-horizon comparisons while also forecasting the next laboratory, medication and diagnosis events. A concept bottleneck exposes named physiological states that clinicians can inspect. [TBD one sentence on the clinician interface and retrospective validation.] One general forecaster can replace a portfolio of bespoke alert models.

Keywords: electronic health records, foundation models, time-to-event forecasting, clinical decision support, interpretability, concept bottleneck models

1 Introduction

[TBD]

2 Results

2.1 Cohorts and the forecasting task

[TBD]

2.2 One model forecasts the next events across families

[TBD]

2.3 Per-event hazard heads match or beat bespoke models

[TBD]

2.4 Risk evolves in time: survival curves at the bedside

[TBD]

2.5 Cross-dataset replication on eICU-CRD

[TBD]

2.6 Interpretable states: what the bottleneck shows and does not

[TBD]

2.7 An agentic interface for clinicians

[TBD]

3 Discussion

[TBD]

4 Methods

4.1 Data sources and ethics

[TBD]

4.2 Tokenization of patient timelines

[TBD]

4.3 Model

[TBD]

4.4 Training objective

[TBD]

4.5 Clinical concepts

[TBD]

4.6 Evaluation of forecasting

[TBD]

4.7 Alert evaluation and bespoke baselines

[TBD]

4.8 Interpretability tests

[TBD]

4.9 Clinician interface and retrospective validation

[TBD]

4.10 Statistical analysis

[TBD]

Data availability. MIMIC-IV and eICU-CRD are available to credentialed researchers through PhysioNet under their data use agreements. [TBD: GEMINI availability statement.] No patient-level data are distributed with this work; the derived aggregate results are in the code repository.

Code availability. All code (extraction specifications, tokenization, model, training, evaluation and the report generator) is available at [TBD repository URL] under [TBD license], at commit [TBD].

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Fig. 1 Overview. a) A patient timeline as an event stream with same-timestamp bundles. b) Tokenization: codes, value bins, time and visit structure. c) The model: backbone, concept bottleneck, next-event, time-to-next-event and per-event hazard heads. d) Outputs at the bedside: forecasts, survival curves, concept states. [TBD]

Fig. 2 Forecasting quality by event family on held-out patients. [TBD]

Fig. 3 Alerts: hazard heads against tuned bespoke gradient-boosted models. AUROC and Brier score by event and horizon on MIMIC-IV and eICU-CRD; calibration. [TBD]

Appendix A Extended Data

[TBD: extended tables (per-family metrics, per-concept AUROC, GBM feature list and chosen hyper-parameters, intervention sweep).]

References

Fig. 4 Risk over time for example stays. [TBD]